

CRITICAL THINKING FIELD GUIDE

Spot the Technique

18 manipulation techniques in energy and science discourse

Most misleading arguments use the same small set of moves. This guide covers the eighteen most common in energy and science discourse, organized into structural fallacies (errors in argument structure) and rhetorical tactics (moves that work on the reader without engaging the argument). Use it to recognize these patterns in what you read. Use it more carefully to recognize them in what you write.

STRUCTURAL FALLACIES

Errors in argument logic

FALSE DICHOTOMY

Presenting two options as if they are the only two.

EXAMPLE *"You can support energy independence, or you can support renewables."*

RESPONSE Ask what other options were edited out of the frame.

STRAWMAN

Misrepresenting the opposing position to make it easier to argue against.

EXAMPLE *"People who want to slow EV mandates want to keep breathing exhaust forever."*

RESPONSE Describe the actual opposing position before engaging it.

AD HOMINEM

Attacking the person instead of engaging the argument.

EXAMPLE *"She has been an activist her whole career, so her research is suspect."*

RESPONSE Ask whether the method, data, and peer review are being addressed.

HASTY GENERALIZATION

Drawing a broad conclusion from a single example.

EXAMPLE *"My cousin's heat pump failed once, so heat pumps do not work in winter."*

RESPONSE Ask what the full dataset shows.

SLIPPERY SLOPE

Predicting a chain of worse consequences without showing the chain is connected.

EXAMPLE *"First gas stoves, next it is every appliance you own."*

RESPONSE Evaluate the specific proposal on its own merits.

CORRELATION-CAUSATION ERROR

Treating two co-occurring things as if one caused the other.

EXAMPLE *"States with renewable mandates have high electricity prices, so renewables cause high prices."*

RESPONSE Ask what other variables might explain both.

FALSE CAUSE

Identifying the wrong cause for an observed outcome.

EXAMPLE *"Germany added renewables, then burned more coal, so renewables failed."*

RESPONSE Ask what else happened at the same time.

RHETORICAL TACTICS

Moves that work without engaging the argument

CHERRY-PICKING

Selecting a real data point that misrepresents the fuller pattern.

EXAMPLE *"The five warmest years on record have all happened in the last decade."*

RESPONSE Ask for the full context before the point.

WHATABOUTISM

Deflecting from the topic by pointing at a different topic.

EXAMPLE *"Why focus on US emissions when China builds coal plants?"*

RESPONSE Note that both topics can be discussed on their own terms.

FAKE EXPERT

Using credentials from one domain to speak authoritatively in another.

EXAMPLE *"A cardiologist says mRNA technology is unsafe."*

RESPONSE Ask whose specialty actually covers the claim.

MANUFACTURED DOUBT

Framing existing evidence as too uncertain to act on.

EXAMPLE *"Research on battery storage is still emerging, so we should wait."*

RESPONSE Ask whether the claim actually argues the other direction or just delays.

MISLEADING CHART

Using visual scale or selection to distort what the data shows.

EXAMPLE *A y-axis that starts at \$80B to make a 15% change look like 300%.*

RESPONSE Check where the y-axis begins. If it does not start at zero, the visual is doing work.

MOVING THE GOALPOSTS

Shifting the standard of evidence whenever the previous standard is met.

EXAMPLE *A rolling sequence of objections that changes each time the prior objection is answered.*

RESPONSE Look for the pattern across time, not any single claim.

APPEAL TO NATURE

Arguing something is good because it is natural, or bad because it is not.

EXAMPLE *"Natural gas is literally natural."*

RESPONSE Ask whether "natural" tracks "good" in this specific case.

APPEAL TO EMOTION

Substituting emotional pressure for evidence.

EXAMPLE *"Can you look your kids in the eye and tell them you stayed silent?"*

RESPONSE Ask what the evidence would be if the emotional pressure were removed.

BANDWAGON

Arguing a claim is true because many or the right people believe it.

EXAMPLE *"Every serious person in the industry knows nuclear is the only answer."*

RESPONSE Ask for the evidence behind the consensus.

EQUIVOCATION

Shifting the meaning of a word mid-argument.

EXAMPLE *"Clean coal is part of clean energy."*

RESPONSE Pin down the specific definition and check whether it holds across the argument.

APPEAL TO TRADITION

Arguing a practice is right because it is how things have been done.

EXAMPLE *"The grid has been built on baseload for a century, that is what works."*

RESPONSE Ask whether the conditions that built the tradition still apply.

FOR YOUR OWN WORK

Recognizing these techniques in what you read is the easy version. Recognizing them in what you write is the hard version. If you communicate for a living on contested topics, treat this list as a self-check. The techniques that feel most persuasive when you catch them in someone else's argument are the same ones that are hardest to catch in your own.

Continue the work. Read The Forge at clarityforigestudios.substack.com or work with us at clarityforigestudios.com